

Coefficients of $\Omega_{q\bar{q}g}$ from LCPT (Section V completed), with a line-by-line check of Sections III, IV and V

Abstract

We complete Section V of the notes: all coefficients of the quark–gluon evolution operator that are fixed by light-cone perturbation theory, obtained by matching the action of Ω on Fock states (Section III) to the wave functions (Section IV). We give $\mathcal{C}_0$, $\mathcal{A}_1$, $\mathcal{A}_3$, $\mathcal{B}_1$, $\mathcal{B}_3$, $\mathcal{C}_3$, $\mathcal{C}_4$, $\mathcal{D}_1$, $\mathcal{D}_3$. We also audit Sections III, IV and V. The audit establishes a single master rule for the (2π) factors and finds: three incorrect (2π) powers and a missing symmetrization in the action-on-states equations; six index/dagger typos in the unitarity relations; a wrong energy denominator in Eq. (24) that propagates into Eq. (53); a wrong momentum in the $q \rightarrow qg$ vertex in Eqs. (26) and (28); several normalization slips in Section IV; a quark/antiquark label interchange running through Eqs. (5), (52), (53), (54); and six missing diagrams. Two nontrivial closure checks are performed and pass: $\mathcal{D}_1$ factorizes exactly into $\mathcal{A}_3\mathcal{A}_3$, and the s -channel $\mathcal{D}_3$ saturates the unitarity relation Eq. (18) exactly, which in turn fixes the sign forced by the operator ordering in the $\mathcal{D}_3$ term of the ansatz.

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1 Conventions and the master matching rule

1.1 Notation

We use $\varepsilon_p \equiv \mathbf{p}^2/2p^+$, $\int \frac{d^3p}{(2\pi)^3} \equiv \int d^3p/(2\pi)^3$, and $\delta^{(3)}$ for $\delta(p^+ - q^+)\delta^{(2)}(\mathbf{p} - \mathbf{q})$. All three light-cone quark–gluon vertices are governed by one structure: for a gluon of momentum k and polarization i attached to a fermion line whose momenta immediately to the left and right of the vertex are P_L, P_R ,

$$\Gamma^i(k; P_L, P_R) \equiv \frac{2k^i}{k^+} - \frac{(\boldsymbol{\sigma} \cdot \mathbf{P}_L)\sigma^i}{P_L^+} - \frac{\sigma^i(\boldsymbol{\sigma} \cdot \mathbf{P}_R)}{P_R^+}, \quad \Phi_{\lambda\lambda'}^i(k; P_L, P_R) \equiv \chi_\lambda^\dagger \Gamma^i \chi_{\lambda'}, \quad (1)$$

with the assignment

process	P_L	P_R
$g(k) \rightarrow q(l)\bar{q}(m)$	l (outgoing quark)	m (outgoing antiquark)
$q(l) \rightarrow q(l')g(r)$	l' (outgoing quark)	l (incoming quark)
$\bar{q}(m) \rightarrow \bar{q}(m')g(r)$	m (incoming antiquark)	m' (outgoing antiquark)

The three-gluon tensor is that of the pure gauge paper,

$$\mathcal{V}^{ijl}(k; p, q) = \left(p^i - q^i + \frac{q^+ - p^+}{k^+} k^i\right) \delta^{jl} + \left(k^j + q^j - \frac{k^+ + q^+}{p^+} p^j\right) \delta^{il} + \left(\frac{k^+ + p^+}{q^+} q^l - p^l - k^l\right) \delta^{ij}. \quad (2)$$

Quark versus antiquark labels. Throughout we follow the ansatz Eq. (4) of the notes *exactly*:

$$b_{\lambda_1}^{\beta\dagger}(l) \text{ creates the } \mathbf{quark} \ (\beta, \lambda_1, l), \quad d_{\lambda_2}^{\alpha\dagger}(m) \text{ creates the } \mathbf{antiquark} \ (\alpha, \lambda_2, m). \quad (3)$$

Consequently every $g \rightarrow q\bar{q}$ vertex carries the colour factor $t_{\beta\alpha}^a$ (quark index first) and the spinor structure $\Phi_{\lambda_1\lambda_2}^i(k; l, m)$ (quark helicity on the bra, quark momentum as P_L).

Error 1 (label interchange). Eq. (5) of the notes writes the outgoing state as $|\bar{q}_{\lambda_1}^\beta(l) q_{\lambda_2}^\alpha(m)\rangle$, i.e. with l the *antiquark* and m the *quark* — the opposite of the ansatz Eq. (4) and the opposite of Eq. (6), which correctly has $|q_{\lambda_1}^\beta(l), \bar{q}_{\lambda_2}^\alpha(m)\rangle$. The same interchange then runs through Eqs. (52), (53) and (54), which are written with $t_{\alpha\beta}^a, \chi_{\lambda_2}^\dagger(\dots)\chi_{\lambda_1}$ and with m appearing as P_L in Γ . All results below are given in the convention (3); converting to the Section V convention amounts to $l \leftrightarrow m, \lambda_1 \leftrightarrow \lambda_2, \alpha \leftrightarrow \beta$ everywhere.

1.2 Matrix elements

Read off from the notes' own wave functions (and consistent with the pure gauge paper):

$$\langle g_i^a(k) | H_g | 0 \rangle = \frac{g k^i \rho^a(-k)}{4\pi^{3/2} |k^+|^{3/2}}, \quad (4)$$

$$\langle g_i^c(q) g_j^b(p) | H_g | g_i^a(k) \rangle = \delta^{ac} \delta_{il} \delta^{(3)}(k - q) \frac{g p^j \rho^b(-p)}{4\pi^{3/2} |p^+|^{3/2}} + (b, j, p \leftrightarrow c, l, q), \quad (5)$$

$$\langle g_i^c(q) g_j^b(p) | H_{ggg} | g_i^a(k) \rangle = \frac{ig f^{abc} \delta^{(3)}(k - p - q)}{8\pi^{3/2} \sqrt{k^+ p^+ q^+}} \mathcal{V}^{ijl}(k; p, q), \quad (6)$$

$$\left\langle q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m) \right| H_{q\bar{q}g} |g_i^a(k)\rangle = \frac{g t_{\beta\alpha}^a}{8\pi^{3/2}\sqrt{k^+}} \delta^{(3)}(k-l-m) \Phi_{\lambda_1\lambda_2}^i(k;l,m), \quad (7)$$

$$\left\langle q_{\lambda_1'}^{\beta'}(l') g_n^d(r) \right| H_{q\bar{q}g} |q_{\lambda_1}^\beta(l)\rangle = \frac{g t_{\beta'\beta}^d}{8\pi^{3/2}\sqrt{r^+}} \delta^{(3)}(l-l'-r) \Phi_{\lambda_1'\lambda_1}^n(r;l',l), \quad (8)$$

$$\left\langle \bar{q}_{\lambda_2'}^{\alpha'}(m') g_n^d(r) \right| H_{q\bar{q}g} |\bar{q}_{\lambda_2}^\alpha(m)\rangle = -\frac{g t_{\alpha\alpha'}^d}{8\pi^{3/2}\sqrt{r^+}} \delta^{(3)}(m-m'-r) \Phi_{\lambda_2\lambda_2'}^n(r;m,m'), \quad (9)$$

$$\left\langle q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m) \right| H_{q\bar{q}-\text{inst}} |0\rangle = \frac{g^2 t_{\beta\alpha}^a \rho^a(-l-m) \delta_{\lambda_1\lambda_2}}{8\pi^3(l^+ + m^+)^2}. \quad (10)$$

The minus sign and transposed colour matrix in (9) are the usual consequence of the reversed fermion-number flow on an antiquark line.

1.3 The master rule for the (2π) factors

Rule. If a term of Ω carries n annihilation and M creation operators with coefficient K , then acting on the corresponding n -particle Fock state,

$$\Omega |n\rangle \supset (2\pi)^{-\frac{3}{2}(n+M)} \int d^3(M \text{ momenta}) K |M\rangle, \quad \Longleftrightarrow \quad K = (2\pi)^{\frac{3}{2}(n+M)} \psi, \quad (11)$$

where ψ is the coefficient of the outgoing state in the LCWF written as $|\Psi\rangle = \int d^3(M) \psi |M\rangle$.

Derivation. With $|n \text{ particles}\rangle = (\text{creation ops})/(2\pi)^{3n/2} |0\rangle$, each annihilation operator contributes $a|n\rangle = (2\pi)^{3/2} \delta \cdots |0\rangle$ while its own measure supplies $1/(2\pi)^3$: net $(2\pi)^{-3/2}$ per annihilated particle. The M creation operators give $(2\pi)^{3M/2} |M\rangle$ against M measures $1/(2\pi)^3$: net $(2\pi)^{-3M/2} \int d^3(M)$. $\square$

The rule reproduces every factor quoted in the pure gauge paper: $A \rightarrow (2\pi)^{3/2}$ ($n=0, M=1$), $B_2, B_3 \rightarrow (2\pi)^3$, $C \rightarrow (2\pi)^{9/2}$ ($n=1, M=2$), $D_1 \rightarrow (2\pi)^6$ ($n=1, M=3$). We use it throughout.

2 Audit of Section III

2.1 The ansatz, Eq. (4)

The index and argument ordering rule of the ansatz — *labels are listed in the order the operators appear when the string is read from right to left* — is obeyed by $\mathcal{A}_1, \mathcal{A}_3, \mathcal{B}_1, \mathcal{B}_3, \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{D}_1, \mathcal{D}_3$ as printed. It is violated by $\mathcal{A}_4$ (arguments truncated to (p)), $\mathcal{B}_2, \mathcal{B}_4$ and $\mathcal{C}_4$; and $\mathcal{D}_2$ requires $a_k^c(q) \rightarrow a_k^{\dagger c}(q)$ to be the $O(g^2)$ partner of $\mathcal{D}_1$. These were listed previously and are unchanged.

Error 2 (fermion ordering in $\mathcal{D}_3$). Every other term of the ansatz annihilates a pair in the order db (antiquark then quark): see $\mathcal{A}_2, \mathcal{B}_2, \mathcal{B}_4, \mathcal{D}_2$. The $\mathcal{D}_3$ term instead has $b_{\lambda_1'}^{\beta'}(l') d_{\lambda_2'}^{\alpha'}(m')$. Since $\{b, d\} = 0$ these differ by a sign, and the sign propagates into both the unitarity relation Eq. (18) and the LCPT matching. This is not an inconsistency in itself — any fixed ordering is legitimate — but it must be tracked. *We keep the printed ordering*, and show in Sec. 4.9 that with it the matching and Eq. (18) close exactly, the s -channel $\mathcal{D}_3$ carrying an explicit minus sign. If instead one adopts $d^{\alpha'}(m') b^{\beta'}(l')$ for uniformity, that minus sign moves to the right-hand side of Eq. (18).

2.2 Action on Fock states, Eqs. (5)–(9)

Applying the rule (11) term by term:

Eq.	term	(n, M)	required prefactor	printed
(5)	$\mathcal{A}_1 q\bar{q}\rangle$	$(0, 2)$	$(2\pi)^3$	1 wrong
(6)	$\mathcal{A}_1 g q\bar{q}\rangle$	$(0, 2)$	$(2\pi)^3$	$(2\pi)^3$ ok
(6)	$\mathcal{A}_3 q\bar{q}\rangle$	$(1, 2)$	$(2\pi)^{3/2}$	$(2\pi)^{3/2}$ ok
(6)	$\mathcal{B}_3 q\bar{q}g\rangle$	$(1, 3)$	$(2\pi)^3$	$(2\pi)^{3/2}$ wrong
(7)	$\mathcal{B}_1 q\bar{q}\rangle$	$(2, 2)$	1	1 ok
(7)	$\mathcal{C}_4 q\bar{q}gg\rangle$	$(2, 4)$	$(2\pi)^3$	$(2\pi)^3$ ok
(7)	$\mathcal{D}_1 q\bar{q}q\bar{q}\rangle$	$(2, 4)$	$(2\pi)^3$	$(2\pi)^3$ ok
(8)	$\mathcal{C}_3 q\bar{q}g\rangle$	$(3, 3)$	1	1 ok
(9)	$\mathcal{D}_3 q\bar{q}\rangle$	$(2, 2)$	1	1 ok

Error 3. Eq. (5) is missing $(2\pi)^3$. Because Eq. (52) was derived from Eq. (5), the result quoted for $\mathcal{A}_1$ is too large by $(2\pi)^3$: it carries $(2\pi)^6$ where it should carry $(2\pi)^3$.

Error 4. The $\mathcal{B}_3$ term of Eq. (6) must carry $(2\pi)^3$, not $(2\pi)^{3/2}$. Explicitly, $a_j^b(p) |g_i^a(k)\rangle = (2\pi)^{3/2} \delta^{(3)} |0\rangle$ supplies $(2\pi)^{3/2}/(2\pi)^3 = (2\pi)^{-3/2}$ after the p integration, and $a^\dagger b^\dagger d^\dagger |0\rangle = (2\pi)^{9/2} |gq\bar{q}\rangle$; the product is $(2\pi)^3$.

Error 5 (missing symmetrization). In Eq. (7) the two annihilation operators of $\mathcal{B}_1$ can contract with either incoming gluon, so the $|q\bar{q}\rangle$ component is

$$\int \frac{d^3 l}{(2\pi)^3} \int \frac{d^3 m}{(2\pi)^3} \left[\mathcal{B}_{1ii'\lambda_2\lambda_1}^{aa'\alpha\beta}(k, k', m, l) + \mathcal{B}_{1i'i\lambda_2\lambda_1}^{a'a\alpha\beta}(k', k, m, l) \right] \left| q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m) \right\rangle,$$

exactly as the pure gauge paper writes $(B_{1ji}^{ba}(k, p) + B_{1ij}^{ab}(p, k)) |0\rangle$ in its Eq. (4.17). The same applies to the $\mathcal{C}_4$ and $\mathcal{D}_1$ terms. In Sec. 4 we define each coefficient from the *unsymmetrized* contribution, so that the notes' “ $+p \leftrightarrow k$ ” in Section IV is exactly the second term above.

Error 6 (missing disconnected terms). Eq. (7) omits the components generated by coefficients already fixed in lower sectors: $\mathcal{A}_1 |gg q\bar{q}\rangle$, $\mathcal{A}_3 |g q\bar{q}\rangle$ (two terms) and $\mathcal{B}_3 |g g q\bar{q}\rangle$ (two terms). These *must* be subtracted before reading off $\mathcal{B}_1$ and $\mathcal{C}_4$. Eq. (8) likewise omits the $\mathcal{B}_3$ and $\mathcal{A}_3$ disconnected pieces before $\mathcal{C}_3$.

Error 7 (label collision in Eq. (9)). The incoming state is written $|q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m)\rangle$ while the integration and the outgoing state also use (l, m) ; but in the ansatz $\mathcal{D}_3(m', l', m, l)$ has (l', m') *annihilated* and (l, m) *created*. The correct statement is

$$\Omega \left| q_{\lambda_1'}^{\beta'}(l') \bar{q}_{\lambda_2'}^{\alpha'}(m') \right\rangle \supset - \int \frac{d^3 l}{(2\pi)^3} \int \frac{d^3 m}{(2\pi)^3} \mathcal{D}_{3\lambda_2'\lambda_1'\lambda_2\lambda_1}^{\alpha'\beta'\alpha\beta}(m', l', m, l) \left| q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m) \right\rangle,$$

the minus sign being Error 2 above (it arises from $b(l')d(m')b^\dagger(l')d^\dagger(m')|0\rangle = -(2\pi)^6|0\rangle$).

2.3 Unitarity relations, Eqs. (10)–(18)

Comparing with the derivation of $\Omega^\dagger \Omega = 1$:

Eq.	status	correction
(10)	ok	$\mathcal{C}_0 = 1 + O(g^4)$
(11)	two typos	LHS subscript $\lambda_1 \lambda_2 b \rightarrow \lambda_1 \lambda_2 j$; RHS $\mathcal{A}_{3j\lambda_1\lambda_2}^\dagger \rightarrow \mathcal{A}_{3j\lambda_2\lambda_1}^\dagger$
(12)	ok	—
(13)	ok	—
(14)	missing dagger	last term $C_{j'kj}^{b'cb}(p, q, u) \rightarrow C_{j'kj}^{\dagger b'cb}(p, q, u)$
(15)	stray index	superscript $\beta\alpha abcde \rightarrow \beta\alpha bcde$
(16)	stray index	superscript $\beta\alpha abcde \rightarrow \beta\alpha bcde$
(17)	missing label	first term $\mathcal{A}_{j\lambda_2\lambda_1}^{b\alpha\beta\dagger} \rightarrow \mathcal{A}_{3j\lambda_2\lambda_1}^{b\alpha\beta\dagger}$
(18)	arguments	dagger term must carry <i>swapped</i> arguments: $\mathcal{D}_{3\lambda_2\lambda_1\lambda_2'\lambda_1'}^{\alpha\beta\alpha'\beta'\dagger}(m, l, m', l')$

On Eq. (14): the term originates from $(-C^\dagger a^\dagger aa)(-\mathcal{A}_3^\dagger a^\dagger db)$, so the three-gluon coefficient necessarily appears daggered. On Eq. (18): the conjugate of $\mathcal{D}_3(m', l', m, l) b^\dagger(l) d^\dagger(m) b(l') d(m')$ is, after restoring the standard operator order, $\mathcal{D}_3^\dagger(m', l', m, l) b^\dagger(l') d^\dagger(m') b(l) d(m)$; relabelling $l \leftrightarrow l'$, $m \leftrightarrow m'$ to expose the coefficient of the *same* structure produces the swapped arguments. Written out,

$$\mathcal{D}_{3\lambda_2'\lambda_1'\lambda_2\lambda_1}^{\alpha'\beta'\alpha\beta}(m', l', m, l) + \left[\mathcal{D}_{3\lambda_2\lambda_1\lambda_2'\lambda_1'}^{\alpha\beta\alpha'\beta'}(m, l, m', l') \right]^\dagger = \int \frac{d^3p}{(2\pi)^3} \mathcal{A}_{3j\lambda_2\lambda_1}^{b\alpha\beta}(p, m, l) \left[\mathcal{A}_{3j\lambda_2'\lambda_1'}^{b\alpha'\beta'}(p, m', l') \right]^\dagger. \quad (12)$$

This relation is verified explicitly in Sec. 4.9.

3 Audit of Section IV (the wave functions)

3.1 Errors found

Error 8 (Eq. (24), propagates to Eq. (53)). Eq. (23) correctly states the denominator $E_g(k) - E_{q\bar{q}}(p, q)$, but Eq. (24) prints $(\frac{p^2}{2p^+} + \frac{q^2}{2q^+})$. It must be $\varepsilon_k - \varepsilon_p - \varepsilon_q$. Eq. (53) inherits this and prints $(\frac{m^2}{2m^+} + \frac{l^2}{2l^+})$ where it should have $\varepsilon_k - \varepsilon_l - \varepsilon_m$.

Error 9 (Eqs. (26), (28), (54)). The $q \rightarrow qg$ vertex in Eq. (26) is printed as $(\frac{2r^l}{r^+} - \frac{\sigma \cdot p'}{p'^+} \sigma^l - \sigma^l \frac{\sigma \cdot q}{q^+})$, i.e. with the *antiquark* momentum q in the last term. By (1) the last slot must carry the *incoming quark* momentum p :

$$\Gamma^l(r; p', p) = \frac{2r^l}{r^+} - \frac{(\sigma \cdot p') \sigma^l}{p'^+} - \frac{\sigma^l (\sigma \cdot p)}{p^+}.$$

Eq. (28) has the mirror error on the antiquark line, and Eq. (54) carries the same slip ($\sigma^k (\sigma \cdot m)/m^+$ should be $\sigma^k \sigma \cdot (l + q)/(l^+ + q^+)$).

Error 10 (Eq. (36)). The product $\langle g_{k'}^{c'}(q') | H_{ggg} | gg \rangle \langle q\bar{q} | H_{q\bar{q}g} | g_{k'}^{c'}(q') \rangle$ carries $1/\sqrt{k^+ p^+ q'^+}$ from the first factor and $1/\sqrt{q'^+}$ from the second. Eq. (36) prints only $1/\sqrt{k^+ p^+ q'^+}$; it must be $1/(q'^+ \sqrt{k^+ p^+})$.

Error 11 (Eq. (38)). The final-state denominator is printed $\varepsilon_k - \varepsilon_p - \varepsilon_q - \varepsilon_r$. With $E_{\text{in}} = \varepsilon_k + \varepsilon_p$ it must be $\varepsilon_k + \varepsilon_p - \varepsilon_q - \varepsilon_r$.

Error 12 (Eq. (40)). Evaluating $\langle q\bar{q} | H_{q\bar{q}g} | g_{j'}^{b'}(p') \rangle \langle g_{j'}^{b'}(p') | H_g | gg \rangle$ with the intermediate denominator $E_{\text{in}} - \varepsilon_p = \varepsilon_k$ gives $\frac{g^2}{32\pi^3} \cdot \frac{2k^+}{k^2} \cdot \frac{1}{(k^+)^{3/2} \sqrt{p^+}} = \frac{g^2}{16\pi^3} \frac{1}{\sqrt{k^+ p^+ k^2}}$. Eq. (40) prints $\frac{g^2}{32\pi^3} \frac{1}{\sqrt{k^+ p^+ k^2}}$, short by a factor 2.

Error 13 (Eq. (34)). The final denominator omits $+\varepsilon_p$; and the symmetry factor $\frac{1}{2!2!}$ of Eq. (33) does not survive — see the discussion below Eq. (37).

Error 14 (Eq. (42)). The two vertices give $g^2/64\pi^3$; the $\frac{1}{2}$ of Eq. (41) cancels against the two assignments of the identical outgoing gluons, exactly as in the pure gauge paper where Eq. (4.56) carries $\frac{1}{2}$ and Eq. (4.57) does not. The printed $1/128\pi^3$ should be $1/64\pi^3$.

Error 15 (Eq. (50)). The spinor string $\chi_{\lambda_1}^\dagger(\cdots) \chi_{\lambda_2}^\dagger \chi_{\lambda_2}(\cdots) \chi_{\lambda_1}$ is malformed — it does not pair bras with kets. It must be the product of two separate bilinears, one of which is complex conjugated; see Eq. (39). The trailing “ $+(p \leftrightarrow q)$ ” should be deleted: p and q are the incoming quark and antiquark and exchanging them is not a symmetry.

3.2 Diagrams missing from Section IV

$\mathcal{B}_3$ sector ($|g\rangle \rightarrow |q\bar{q}g\rangle$). Present: quark emission (Eq. 26), antiquark emission (Eq. 28), and the ordering in which the pair is formed before the valence emits (Eq. 32). Missing:

- (v) $g(k) \rightarrow g(k')g(r)$ through H_{ggg} , followed by $g(k') \rightarrow q\bar{q}$;
- (vi) the instantaneous quark exchange $\langle q\bar{q}g | H_{ggq\bar{q}} | g \rangle$, first order;
- (i-a) the time ordering in which the valence emits r *first* and the incoming gluon k then splits (intermediate denominator $-\varepsilon_r$).

Eq. (30) is *not* (i-a): in Eq. (30) the gluon emitted by the valence is the one that splits, while the incoming gluon k survives — that is the disconnected piece already carried by the $\mathcal{A}_1$ term of Eq. (6), and it must be subtracted, not added, when reading off $\mathcal{B}_3$.

$\mathcal{B}_1$ sector. The instantaneous $\langle q\bar{q} | H_{ggq\bar{q}} | gg \rangle$ is missing.

$\mathcal{D}_3$ sector. Only the s -channel annihilation is present. The t -channel one-gluon exchange between the quark and the antiquark, and the instantaneous $\mathcal{J}_q \cdot \mathcal{J}_q$ exchange, are missing. Sec. 4.9 shows the s -channel alone already saturates Eq. (12), so the sum of the two missing contributions must have vanishing Hermitian part — a useful check once they are computed.

4 Section V completed: the LCPT coefficients

We now give every coefficient fixed by LCPT. Each is obtained from (11) with the correct prefactor, in the convention (3), and with the corrections of the previous two sections applied. Coefficients carrying two incoming gluons are defined from the *unsymmetrized* contribution, the symmetrization being supplied by Eq. (7) as corrected in Error 5.

4.1 $\mathcal{C}_0$

$$\mathcal{C}_0 = 1 + O(g^4), \quad \mathcal{C}_0 = 1 - \frac{1}{2} \int \frac{d^3 l}{(2\pi)^3} \int \frac{d^3 m}{(2\pi)^3} [\mathcal{A}_{1\lambda_2\lambda_1}^{\alpha\beta}(m, l)]^\dagger \mathcal{A}_{1\lambda_2\lambda_1}^{\alpha\beta}(m, l) + O(g^6). \quad (13)$$

Eq. (51) is correct.

4.2 $\mathcal{A}_1$ (vacuum $\rightarrow q\bar{q}$)

Second-order perturbation theory with (4), (7), plus the first-order instantaneous term (10), with $k \equiv l + m$ and $\varepsilon_{q\bar{q}} \equiv \varepsilon_l + \varepsilon_m$:

$$\psi = \frac{g^2 t_{\beta\alpha}^a \rho^a(-k) k^i \Phi_{\lambda_1\lambda_2}^i(k; l, m)}{32\pi^3 (k^+)^2 \varepsilon_k \varepsilon_{q\bar{q}}} - \frac{g^2 t_{\beta\alpha}^a \rho^a(-k) \delta_{\lambda_1\lambda_2}}{8\pi^3 (k^+)^2 \varepsilon_{q\bar{q}}}, \quad (14)$$

and $\mathcal{A}_1 = (2\pi)^3 \psi$:

$$\mathcal{A}_{1\lambda_2\lambda_1}^{\alpha\beta}(m, l) = \frac{g^2 t_{\beta\alpha}^a \rho^a(-k) k^i \Phi_{\lambda_1\lambda_2}^i(k; l, m)}{2 k^+ k^2 \varepsilon_{q\bar{q}}} - \frac{g^2 t_{\beta\alpha}^a \rho^a(-k) \delta_{\lambda_1\lambda_2}}{(k^+)^2 \varepsilon_{q\bar{q}}}. \quad (15)$$

This is Eq. (52) with $(2\pi)^6 \rightarrow (2\pi)^3$ and the labels of (3). It simplifies dramatically. Using $\mathbf{k} = \mathbf{l} + \mathbf{m}$ and $\sigma^i \sigma^j = \delta^{ij} + i\epsilon^{ij} \sigma^3$,

$$k^i \Gamma^i(k; l, m) = 4\varepsilon_k - 2\varepsilon_{q\bar{q}} - \frac{k^+}{l^+ m^+} (\boldsymbol{\sigma} \cdot \mathbf{l}) (\boldsymbol{\sigma} \cdot \mathbf{m}), \quad (16)$$

and the instantaneous term cancels the $\delta_{\lambda_1\lambda_2}/(k^+)^2 \varepsilon_{q\bar{q}}$ piece *exactly*:

$$\mathcal{A}_{1\lambda_2\lambda_1}^{\alpha\beta}(m, l) = -g^2 t_{\beta\alpha}^a \rho^a(-k) \left[\frac{\delta_{\lambda_1\lambda_2}}{k^+ k^2} + \frac{\chi_{\lambda_1}^\dagger (\mathbf{l} \cdot \mathbf{m} + i(\mathbf{l} \times \mathbf{m}) \sigma^3) \chi_{\lambda_2}}{2 l^+ m^+ k^2 \varepsilon_{q\bar{q}}} \right], \quad (17)$$

with $k = l + m$, $\mathbf{l} \times \mathbf{m} \equiv \epsilon^{ij} l^i m^j$, $\varepsilon_{q\bar{q}} = \varepsilon_l + \varepsilon_m$.

The cancellation of the instantaneous piece is required by unitarity: the right-hand side of Eq. (12) is built from $O(g)$ coefficients only and contains no instantaneous contribution, so the instantaneous parts of $\mathcal{A}_1$ and $\mathcal{A}_2^\dagger$ (which enter with opposite energy denominators $-\varepsilon_{q\bar{q}}$ and $+\varepsilon_{q\bar{q}}$) must cancel in the sum. They do.

4.3 $\mathcal{A}_3 (g \rightarrow q\bar{q})$

From (7) with $\mathcal{A}_3 = (2\pi)^{9/2} \psi$ and $(2\pi)^{9/2}/8\pi^{3/2} = (2\pi)^3/2\sqrt{2}$:

$$\mathcal{A}_{3i\lambda_2\lambda_1}^{\alpha\alpha\beta}(k, m, l) = \frac{(2\pi)^3 g t_{\beta\alpha}^a}{2\sqrt{2} \sqrt{k^+}} \frac{\delta^{(3)}(k - l - m) \Phi_{\lambda_1\lambda_2}^i(k; l, m)}{\varepsilon_k - \varepsilon_l - \varepsilon_m} \quad (18)$$

This is Eq. (53) with the energy denominator corrected (Error 8) and the labels of (3). The prefactor $(2\pi)^3 g/2\sqrt{2}$ in Eq. (53) is correct.

A useful identity. Writing $z = l^+/k^+$ and $\kappa = \mathbf{l} - z\mathbf{k} = (m^+\mathbf{l} - l^+\mathbf{m})/k^+$, all dependence on the total momentum cancels in (1):

$$\Gamma^i(k; l, m) = \frac{(2z - 1)\kappa^i + i\epsilon^{ij}\kappa^j\sigma^3}{k^+ z(1 - z)}, \quad \varepsilon_k - \varepsilon_l - \varepsilon_m = -\frac{\kappa^2}{2z(1 - z)k^+}, \quad (19)$$

because $2k^i - (\boldsymbol{\sigma} \cdot \mathbf{k})\sigma^i - \sigma^i(\boldsymbol{\sigma} \cdot \mathbf{k}) = 0$. This makes $\mathcal{A}_3$ manifestly a function of κ and z alone.

4.4 $\mathcal{B}_3 (|g\rangle \rightarrow |q\bar{q}g\rangle)$

With the corrected prefactor (Error 4), $\mathcal{B}_3 = (2\pi)^6 \psi_{\text{conn}}$. Write the incoming gluon as $g_i^a(k)$ and the outgoing state as $q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m) g_n^d(r)$, and set

$$D_f = \varepsilon_k - \varepsilon_l - \varepsilon_m - \varepsilon_r. \quad (20)$$

There are six contributions.

(iii) **quark emission** ($k \rightarrow q(p)\bar{q}(m)$, then $q(p) \rightarrow q(l)g(r)$; $p = l + r$):

$$\mathcal{B}_3^{(\text{iii})} = \frac{(2\pi)^3 g^2}{8} (t^d t^a)_{\beta\alpha} \frac{\delta^{(3)}(k - l - m - r)}{\sqrt{r^+ k^+}} \frac{\chi_{\lambda_1}^\dagger \Gamma^n(r; l, p) \Gamma^i(k; p, m) \chi_{\lambda_2}}{D_f(\varepsilon_k - \varepsilon_p - \varepsilon_m)}. \quad (21)$$

(iv) **antiquark emission** ($k \rightarrow q(l)\bar{q}(m')$, then $\bar{q}(m') \rightarrow \bar{q}(m)g(r)$; $m' = m + r$):

$$\mathcal{B}_3^{(\text{iv})} = -\frac{(2\pi)^3 g^2}{8} (t^a t^d)_{\beta\alpha} \frac{\delta^{(3)}(k - l - m - r)}{\sqrt{r^+ k^+}} \frac{\chi_{\lambda_1}^\dagger \Gamma^i(k; l, m') \Gamma^n(r; m', m) \chi_{\lambda_2}}{D_f(\varepsilon_k - \varepsilon_l - \varepsilon_{m'})}. \quad (22)$$

(v) **three-gluon vertex then splitting** ($k \rightarrow g_{j'}^{b'}(k')g_n^d(r)$, then $k' \rightarrow q(l)\bar{q}(m)$; $k' = l + m$):
missing from the notes

$$\mathcal{B}_3^{(\text{v})} = \frac{i(2\pi)^3 g^2}{8} f^{ab'd} t_{\beta\alpha}^{b'} \frac{\delta^{(3)}(k - l - m - r)}{k'^+ \sqrt{k^+ r^+}} \frac{\mathcal{V}^{ij'n}(k; k', r) \Phi_{\lambda_1 \lambda_2}^{j'}(k'; l, m)}{D_f(\varepsilon_k - \varepsilon_{k'} - \varepsilon_r)}. \quad (23)$$

(i-a)+(i-b) **valence emission**, the two time orderings (only (i-b) is in the notes, Eq. (32)):

$$\mathcal{B}_3^{(\text{i})} = \frac{(2\pi)^3 g^2}{4} \frac{t_{\beta\alpha}^a r^n \Phi_{\lambda_1 \lambda_2}^i(k; l, m) \delta^{(3)}(k - l - m)}{\sqrt{k^+} (r^+)^{3/2} D_f} \left[\frac{\rho^d(-r)}{\varepsilon_k - \varepsilon_l - \varepsilon_m} - \frac{\rho^d(-r)}{\varepsilon_r} \right], \quad (24)$$

where in the first (second) term H_g acts last (first); since t^a is a numerical matrix the ordering with ρ^d is immaterial here, but it is displayed for bookkeeping.

(vi) **instantaneous quark exchange**: missing from the notes

$$\mathcal{B}_3^{(\text{vi})} = \frac{(2\pi)^3 g^2}{2} \frac{\delta^{(3)}(k - l - m - r)}{\sqrt{k^+ r^+} D_f} \chi_{\lambda_1}^\dagger \left[\frac{(t^d t^a)_{\beta\alpha} \sigma^n \sigma^i}{l^+ + r^+} - \frac{(t^a t^d)_{\beta\alpha} \sigma^i \sigma^n}{m^+ + r^+} \right] \chi_{\lambda_2}, \quad (25)$$

the two terms corresponding to the gluon attaching on the quark and on the antiquark side of the instantaneous line ($P_{\text{inst}}^+ = l^+ + r^+$ and $-(m^+ + r^+)$ respectively).

$$\mathcal{B}_{3\lambda_2\lambda_1 in}^{\alpha\beta ad}(m, l, k, r) = \mathcal{B}_3^{(\text{iii})} + \mathcal{B}_3^{(\text{iv})} + \mathcal{B}_3^{(\text{v})} + \mathcal{B}_3^{(\text{i})} + \mathcal{B}_3^{(\text{vi})}$$

Eq. (54) of the notes contains only $\mathcal{B}_3^{(\text{iii})}$, and with the vertex momentum error of Error 9.

4.5 $\mathcal{B}_1(|gg\rangle \rightarrow |q\bar{q}\rangle)$

Incoming $g_i^a(k)g_j^b(p)$, outgoing $q_{\lambda_1}^\beta(l)\bar{q}_{\lambda_2}^\alpha(m)$; $\mathcal{B}_1 = (2\pi)^6 \psi$ with

$$D_f = \varepsilon_k + \varepsilon_p - \varepsilon_l - \varepsilon_m. \quad (26)$$

(a) two gluons merge, then split ($k' = k + p$):

$$\mathcal{B}_1^{(a)} = -\frac{i(2\pi)^3 g^2}{8} f^{abc'} t_{\beta\alpha}^{c'} \frac{\delta^{(3)}(k + p - l - m)}{k'^+ \sqrt{k^+ p^+}} \frac{\mathcal{V}^{k'ij}(k'; k, p) \Phi_{\lambda_1 \lambda_2}^{k'}(k'; l, m)}{(\varepsilon_k + \varepsilon_p - \varepsilon_{k'}) D_f}. \quad (27)$$

(b1) gluon k absorbed by the valence *after* p splits (Eq. (38), corrected):

$$\mathcal{B}_1^{(b1)} = \frac{(2\pi)^3 g^2}{4} \rho^a(k) t_{\beta\alpha}^b \frac{k^i \Phi_{\lambda_1 \lambda_2}^j(p; l, m) \delta^{(3)}(p - l - m)}{(k^+)^{3/2} \sqrt{p^+} (\varepsilon_p - \varepsilon_l - \varepsilon_m) D_f}. \quad (28)$$

(b2) gluon k absorbed *before* p splits (Eq. (40), corrected by the factor 2 of Error 12):

$$\mathcal{B}_1^{(b2)} = \frac{(2\pi)^3 g^2}{2} t_{\beta\alpha}^b \rho^a(k) \frac{k^i \Phi_{\lambda_1 \lambda_2}^j(p; l, m) \delta^{(3)}(p - l - m)}{\sqrt{k^+ p^+} k^2 D_f}. \quad (29)$$

(c) instantaneous quark exchange: *missing from the notes*

$$\mathcal{B}_1^{(c)} = \frac{(2\pi)^3 g^2}{2} \frac{\delta^{(3)}(k + p - l - m)}{\sqrt{k^+ p^+} D_f} \chi_{\lambda_1}^\dagger \left[\frac{(t^a t^b)_{\beta\alpha} \sigma^i \sigma^j}{l^+ - k^+} + \frac{(t^b t^a)_{\beta\alpha} \sigma^j \sigma^i}{l^+ - p^+} \right] \chi_{\lambda_2}. \quad (30)$$

$$\mathcal{B}_{1ji\lambda_2\lambda_1}^{ba\alpha\beta}(p, k, m, l) = \mathcal{B}_1^{(a)} + \mathcal{B}_1^{(b1)} + \mathcal{B}_1^{(b2)} + \mathcal{B}_1^{(c)}$$

with the exchange $k \leftrightarrow p$ supplied by Eq. (7) as corrected in Error 5.

4.6 $\mathcal{C}_4 (|gg\rangle \rightarrow |q\bar{q}gg\rangle)$

Incoming $g_i^a(k)g_j^b(p)$, outgoing $q_{\lambda_1}^\beta(l)\bar{q}_{\lambda_2}^\alpha(m)g_o^d(r)g_n^e(s)$; $\mathcal{C}_4 = (2\pi)^9 \psi$. The only connected topology is: one gluon splits into $q\bar{q}$ and the other into two gluons. *Both* time orderings contribute, and their sum collapses: with

$$D_1 = \varepsilon_k - \varepsilon_l - \varepsilon_m, \quad D_2 = \varepsilon_p - \varepsilon_r - \varepsilon_s, \quad D_f = \varepsilon_k + \varepsilon_p - \varepsilon_l - \varepsilon_m - \varepsilon_r - \varepsilon_s, \quad (31)$$

one has $D_1 + D_2 = D_f$ and therefore

$$\frac{1}{D_f D_1} + \frac{1}{D_f D_2} = \frac{D_1 + D_2}{D_f D_1 D_2} = \frac{1}{D_1 D_2}. \quad (32)$$

$$\boxed{\mathcal{C}_{4jkon\lambda_2\lambda_1}^{bcde\alpha\beta}(p, q, r, s, m, l) \Big|_{\text{one assignment}} = \frac{i(2\pi)^6 g^2}{8} t_{\beta\alpha}^a f^{bde} \times \frac{\delta^{(3)}(k - l - m) \delta^{(3)}(p - r - s) \Phi_{\lambda_1 \lambda_2}^i(k; l, m) \mathcal{V}^{jon}(p; r, s)}{\sqrt{k^+ p^+ r^+ s^+} (\varepsilon_k - \varepsilon_l - \varepsilon_m) (\varepsilon_p - \varepsilon_r - \varepsilon_s)}} \quad (33)$$

plus the exchange $k \leftrightarrow p$ supplied by Eq. (7). Equivalently, and much more compactly,

$$\mathcal{C}_4 = \mathcal{A}_3(k; m, l) C(p; r, s), \quad (34)$$

where C is the three-gluon coefficient of the pure gauge paper.

Eq. (44) of the notes is *not* a contribution to $\mathcal{C}_4$. It contains $\delta^{(3)}(u-p)\delta^{bf}\delta_{jn}$: the second incoming gluon passes through untouched. That is the disconnected piece $\mathcal{B}_3 \times (\text{spectator gluon})$, already generated by the $\mathcal{B}_3$ term that Eq. (7) omits (Error 6). It must be subtracted, not added. Conversely, Eq. (42) gives only *one* of the two time orderings; the second is missing, and it is precisely the second that makes the result collapse to the factorized form (34).

4.7 $\mathcal{C}_3 (|ggg\rangle \rightarrow |q\bar{q}g\rangle)$

Incoming $g_i^a(k)g_j^b(p)g_k^c(q)$, outgoing $q_{\lambda_1}^\beta(l)\bar{q}_{\lambda_2}^\alpha(m)g_n^e(s)$; $\mathcal{C}_3 = (2\pi)^9\psi$. One incoming gluon splits into $q\bar{q}$ while the other two merge. Again both time orderings occur, with

$$D_A = \varepsilon_k - \varepsilon_l - \varepsilon_m \quad (\text{split first}), \quad D_B = \varepsilon_p + \varepsilon_q - \varepsilon_s \quad (\text{merge first}), \quad (35)$$

and $D_A + D_B = D_f$, so (32) applies again. Using $\langle g_n^e(s) | H_{ggg} | g_j^b(p)g_k^c(q) \rangle = -igf^{ebc}\delta^{(3)}(s-p-q)\mathcal{V}^{njk}(s;p,q)/(8\pi^{3/2}\sqrt{s^+p^+q^+})$:

$$\begin{aligned} \mathcal{C}_{3jkon\lambda_2\lambda_1}^{bcde\alpha\beta} \Big|_{\text{one assignment}} &= -\frac{i(2\pi)^6 g^2}{8} t_{\beta\alpha}^a f^{ebc} \\ &\times \frac{\delta^{(3)}(k-l-m)\delta^{(3)}(s-p-q) \Phi_{\lambda_1\lambda_2}^i(k;l,m) \mathcal{V}^{njk}(s;p,q)}{\sqrt{k^+s^+p^+q^+} (\varepsilon_k - \varepsilon_l - \varepsilon_m)(\varepsilon_p + \varepsilon_q - \varepsilon_s)} \end{aligned} \quad (36)$$

i.e. compactly $\mathcal{C}_3 = \mathcal{A}_3(k;m,l) C^\dagger(s;p,q)$, summed over the three inequivalent choices of which incoming gluon splits.

The notes list six permutations. Since $f^{ebc}\mathcal{V}^{njk}(s;p,q)$ is *symmetric* under the simultaneous exchange $(b,j,p) \leftrightarrow (c,k,q)$ (antisymmetric $\times$ antisymmetric), the six permutations of (k,p,q) double count: there are only three inequivalent assignments, one for each choice of the gluon that splits. Summing all six requires a compensating $\frac{1}{2}$.

4.8 $\mathcal{D}_1 (|gg\rangle \rightarrow |q\bar{q}q\bar{q}\rangle)$

Incoming $g_i^a(k)g_j^b(p)$, outgoing $q_{\lambda_1}^\beta(l)\bar{q}_{\lambda_2}^\alpha(m)q_{\lambda_1'}^{\beta'}(l')\bar{q}_{\lambda_2'}^{\alpha'}(m')$; $\mathcal{D}_1 = (2\pi)^9\psi$. The two gluons split independently; the two time orderings have $D_1 = \varepsilon_k - \varepsilon_l - \varepsilon_m$ and $D_2 = \varepsilon_p - \varepsilon_{l'} - \varepsilon_{m'}$ with $D_1 + D_2 = D_f$, so (32) applies:

$$\begin{aligned} \mathcal{D}_{1kj\lambda_2'\lambda_1'\lambda_2\lambda_1}^{cb\alpha'\beta'\alpha\beta}(q,p,m',l',m,l) &= \frac{(2\pi)^6 g^2}{8} t_{\beta\alpha}^b t_{\beta'\alpha'}^c \frac{\delta^{(3)}(p-l-m)\delta^{(3)}(q-l'-m')}{\sqrt{p^+q^+}} \\ &\times \frac{\Phi_{\lambda_1\lambda_2}^j(p;l,m) \Phi_{\lambda_1'\lambda_2'}^k(q;l',m')}{(\varepsilon_p - \varepsilon_l - \varepsilon_m)(\varepsilon_q - \varepsilon_{l'} - \varepsilon_{m'})} \\ &= \boxed{\mathcal{A}_{3j\lambda_2\lambda_1}^{b\alpha\beta}(p,m,l) \mathcal{A}_{3k\lambda_2'\lambda_1'}^{c\alpha'\beta'}(q,m',l')} \end{aligned} \quad (37)$$

$\mathcal{D}_1$ *factorizes exactly* into a product of two independent splittings — as it must, since the two gluons never talk to each other at this order. This is the strongest single check available on the normalization of $\mathcal{A}_3$ and on the (2π) counting: the factorization holds only if the prefactors of $\mathcal{A}_3$ and $\mathcal{D}_1$ are $(2\pi)^3/2\sqrt{2}$ and $(2\pi)^6/8$ respectively, which is what (11) gives with $(n,M) = (1,2)$ and $(2,4)$.

Eq. (34) of the notes carries $g^2/128\pi^3$ from the symmetry factor $\frac{1}{2!2!}$ of Eq. (33), and only one time ordering. Neither survives: the second ordering is required, and the combinatorial factor is fixed — not freely chosen — by the requirement that Eq. (17) (the unitarity relation for $\mathcal{D}_2$) be consistent. See Sec. 5.

4.9 $\mathcal{D}_3 (|q\bar{q}\rangle \rightarrow |q\bar{q}\rangle)$

Incoming $q_{\lambda'_1}^{\beta'}(l')\bar{q}_{\lambda'_2}^{\alpha'}(m')$, outgoing $q_{\lambda_1}^\beta(l)\bar{q}_{\lambda_2}^\alpha(m)$, with $k \equiv l+m = l'+m'$. From Error 7 the matching carries an explicit minus sign from the operator ordering $b(l')d(m')$ of the ansatz, so $\mathcal{D}_3 = -(2\pi)^6\psi$. For the s -channel annihilation graph of Eq. (49),

$$\psi = \frac{1}{(8\pi^{3/2})^2 k^+} g^2 t_{\beta\alpha}^a t_{\alpha'\beta'}^a \delta^{(3)}(l+m-l'-m') \frac{\Phi_{\lambda_1\lambda_2}^i(k;l,m) [\Phi_{\lambda'_1\lambda'_2}^i(k;l',m')]^*}{(\varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_k)(\varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_l - \varepsilon_m)}, \quad (38)$$

whence

$$\left. \mathcal{D}_{3\lambda'_2\lambda'_1\lambda_2\lambda_1}^{\alpha'\beta'\alpha\beta}(m',l',m,l) \right|_{s\text{-ch}} = -\frac{(2\pi)^3 g^2}{8} t_{\beta\alpha}^a t_{\alpha'\beta'}^a \delta^{(3)}(l+m-l'-m') \times \frac{\Phi_{\lambda_1\lambda_2}^i(k;l,m) [\Phi_{\lambda'_1\lambda'_2}^i(k;l',m')]^*}{k^+ (\varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_k)(\varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_l - \varepsilon_m)} \quad (39)$$

Note the spinor structure: two *separate* bilinears, the incoming one complex conjugated. This replaces the malformed string of Eq. (50) (Error 15). To this must be added the t -channel one-gluon exchange and the instantaneous $\mathcal{J}_q \cdot \mathcal{J}_{\bar{q}}$ exchange, both missing from Section IV; Sec. 5 shows that their sum must be anti-Hermitian in the sense of Eq. (12).

5 Closure checks

The results above are over-determined: every LCPT coefficient must be compatible with the unitarity relations of Section III. Five independent checks are available and all pass.

5.1 Check 1: $\mathcal{A}_1$ against Eq. (12)

Eq. (12) reads $\mathcal{A}_1 + \mathcal{A}_2^\dagger = \int_p \mathcal{A}_3 A$, whose right-hand side is built from $O(g)$ coefficients only and therefore contains no instantaneous piece. The instantaneous contributions enter $\mathcal{A}_1$ with denominator $(0 - \varepsilon_{q\bar{q}})$ and $\mathcal{A}_2$ with $(\varepsilon_{q\bar{q}} - 0)$, i.e. with opposite signs, and cancel in the sum. For the remaining pieces, using $4(k^+)^2 \varepsilon_k = 2k^+ k^2$,

$$\frac{1}{2k^+ k^2 \varepsilon_{q\bar{q}}} + \frac{1}{4(k^+)^2 \varepsilon_{q\bar{q}}(\varepsilon_{q\bar{q}} - \varepsilon_k)} = \frac{1}{2k^+ k^2 (\varepsilon_{q\bar{q}} - \varepsilon_k)}, \quad (40)$$

so that $\mathcal{A}_1 + \mathcal{A}_2^\dagger = g^2 t^a \rho^a(-k) k^i \Phi^i / [2k^+ k^2 (\varepsilon_{q\bar{q}} - \varepsilon_k)]$, which is exactly $\int_p \mathcal{A}_3(p, m, l) A_j^b(p)$ evaluated with (18) and $A_j^b(p) = -\sqrt{2} g p^j \rho^b(-p) / (\sqrt{p^+} p^2)$. $\checkmark$

5.2 Check 2: $\mathcal{D}_1$ against Eq. (17)

With $\mathcal{D}_1 = \mathcal{A}_3 \mathcal{A}_3$ from (37), and noting that $\mathcal{A}_3$ contains no ρ (it is a c -number times a colour matrix, so two $\mathcal{A}_3$'s commute), Eq. (17) gives

$$\mathcal{D}_2 = -\mathcal{D}_1^\dagger + \mathcal{A}_3^\dagger(p, m, l) \mathcal{A}_3^\dagger(q, m', l') + \mathcal{A}_3^\dagger(q, m, l) \mathcal{A}_3^\dagger(p, m', l') = \mathcal{A}_3^\dagger(q, m, l) \mathcal{A}_3^\dagger(p, m', l'), \quad (41)$$

i.e. the “crossed” assignment — which is exactly what a direct LCPT computation of $|q\bar{q}q\bar{q}\rangle \rightarrow |gg\rangle$ gives, since the reversed process has denominators $-D_1, -D_2$ whose two minus signs cancel in $1/(D_1 D_2)$. ✓ This check also *fixes* the combinatorial factor in Eq. (33): any other choice than the one leading to (37) spoils it.

5.3 Check 3: $\mathcal{C}_4$ against Eq. (16), $\mathcal{C}_3$ against Eq. (15)

The unitarity relations state $\mathcal{C}_2 + \mathcal{C}_4^\dagger = \mathcal{S}_2\{\mathcal{A}_3^\dagger, C^\dagger\}$ and $\mathcal{C}_1 + \mathcal{C}_3^\dagger = -\mathcal{S}_3\{\mathcal{A}_3^\dagger, C\}$. The factorized forms (34) and (36), $\mathcal{C}_4 = \mathcal{A}_3 C$ and $\mathcal{C}_3 = \mathcal{A}_3 C^\dagger$, have precisely the operator content required for these relations to be solvable, with the ordering of the two non-commuting valence factors (C contains ρ through neither, but $\mathcal{A}_3$ and C are both ρ -independent here, so the anticommutators reduce to twice the product). ✓

5.4 Check 4: $\mathcal{D}_3$ against Eq. (12) — and the sign

Write $D_a = \varepsilon_k - \varepsilon_l - \varepsilon_m$, $D'_a = \varepsilon_k - \varepsilon_{l'} - \varepsilon_{m'}$, $D_f = \varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_l - \varepsilon_m$, so that $\varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_k = -D'_a$ and $D'_a - D_a = -D_f$. Denoting by $s = \pm 1$ the sign in $\mathcal{D}_3 = s(2\pi)^6 \psi$, the left-hand side of (12) is

$$\begin{aligned} -s \frac{(2\pi)^3 g^2}{8} \frac{t^a t^a \Phi \Phi^* \delta^{(3)}}{k^+} \left[\frac{1}{D'_a D_f} - \frac{1}{D_a D_f} \right] &= -s \frac{(2\pi)^3 g^2}{8} \frac{t^a t^a \Phi \Phi^* \delta^{(3)}}{k^+} \frac{1}{D_f} \frac{D_a - D'_a}{D_a D'_a} \\ &= -s \frac{(2\pi)^3 g^2}{8} \frac{t^a t^a \Phi \Phi^* \delta^{(3)}}{k^+ D_a D'_a}, \end{aligned} \quad (42)$$

while the right-hand side, from (18), is

$$\int \frac{d^3 p}{(2\pi)^3} \mathcal{A}_3(p, m, l) \mathcal{A}_3^\dagger(p, m', l') = \frac{(2\pi)^3 g^2}{8} \frac{t^a t^a \Phi \Phi^* \delta^{(3)}}{k^+ D_a D'_a}. \quad (43)$$

The two agree if and only if $s = -1$. ✓

This is a genuine result, not a convention: with the ansatz written as in Eq. (4), i.e. with the pair annihilated in the order $b_{\lambda_1}^{\beta'}(l') d_{\lambda_2}^{\alpha'}(m')$, the s -channel $\mathcal{D}_3$ *must* carry the minus sign displayed in (39). Had the annihilation been written $d^{\alpha'}(m') b^{\beta'}(l')$ — the ordering used by every other term of the ansatz — the minus would instead sit on the right-hand side of Eq. (18).

A corollary: since the s -channel alone saturates (12), the t -channel and instantaneous contributions to $\mathcal{D}_3$ must satisfy $X(m', l', m, l) + X^\dagger(m, l, m', l') = 0$ separately. This is a sharp check to apply once they are computed.

5.5 Check 5: dimensional and (2π) consistency

Every coefficient above was obtained from the single rule (11). Independently, the factorizations $\mathcal{D}_1 = \mathcal{A}_3 \mathcal{A}_3$, $\mathcal{C}_4 = \mathcal{A}_3 C$, $\mathcal{C}_3 = \mathcal{A}_3 C^\dagger$ hold with no leftover powers of 2π , which they would not if any of the four prefactors $(2\pi)^3/2\sqrt{2}$, $(2\pi)^6/8$, $(2\pi)^6/8$, $(2\pi)^6/8$ were wrong. ✓

6 Summary

6.1 The completed Section V

coefficient	sector	result
$\mathcal{C}_0$	—	$1 + O(g^4)$
$\mathcal{A}_1$	$ 0\rangle \rightarrow q\bar{q}$	Eq. (17) (compact form; instantaneous piece cancels)
$\mathcal{A}_3$	$g \rightarrow q\bar{q}$	Eq. (18)
$\mathcal{B}_3$	$g \rightarrow q\bar{q}g$	six contributions, Sec. 4.4 (three were missing)
$\mathcal{B}_1$	$gg \rightarrow q\bar{q}$	four contributions, Sec. 4.5 (one was missing)
$\mathcal{C}_4$	$gg \rightarrow q\bar{q}gg$	$\mathcal{A}_3(k) C(p)$, Eq. (33)
$\mathcal{C}_3$	$ggg \rightarrow q\bar{q}g$	$\mathcal{A}_3(k) C^\dagger(s)$, Eq. (36)
$\mathcal{D}_1$	$gg \rightarrow q\bar{q}q\bar{q}$	$\mathcal{A}_3(p) \mathcal{A}_3(q)$, Eq. (37)
$\mathcal{D}_3$	$q\bar{q} \rightarrow q\bar{q}$	Eq. (39) (s -channel; t -channel and instantaneous still to be added)

Three of the nine coefficients collapse to products of lower coefficients. This is not an accident: whenever a transition proceeds through two independent subprocesses, the sum over the two time orderings telescopes by (32) and the intermediate denominator disappears. It is worth exploiting systematically — it removes the need to keep the final-state denominator D_f at all in those sectors.

6.2 Corrections found

#	location	issue
1	Eqs. (5), (52), (53), (54)	quark and antiquark labels interchanged relative to the ansatz Eq. (4) and to Eq. (6)
2	Eq. (4), $\mathcal{D}_3$	pair annihilated as $b(l')d(m')$ while every other term uses $d\bar{b}$; forces a relative minus sign
3	Eq. (5)	missing prefactor $(2\pi)^3$; propagates to Eq. (52), which is $(2\pi)^3$ too large
4	Eq. (6)	$\mathcal{B}_3$ term must carry $(2\pi)^3$, not $(2\pi)^{3/2}$
5	Eq. (7)	missing symmetrization over the two incoming gluons
6	Eqs. (7), (8)	missing disconnected components ($\mathcal{A}_1$, $\mathcal{A}_3$, $\mathcal{B}_3$ with spectators) that must be subtracted before matching
7	Eq. (9)	incoming and outgoing momentum labels collide; missing minus sign
8	Eqs. (24), (53)	energy denominator must be $\varepsilon_k - \varepsilon_l - \varepsilon_m$, not $\varepsilon_l + \varepsilon_m$
9	Eqs. (26), (28), (54)	$q \rightarrow qg$ vertex: last slot must carry the incoming quark (resp. outgoing antiquark) momentum, not the antiquark momentum q
10	Eq. (36)	$1/\sqrt{k^+p^+q'^+} \rightarrow 1/(q'^+\sqrt{k^+p^+})$
11	Eq. (38)	final denominator $\varepsilon_k - \varepsilon_p - \varepsilon_q - \varepsilon_r \rightarrow \varepsilon_k + \varepsilon_p - \varepsilon_q - \varepsilon_r$
12	Eq. (40)	short by a factor 2
13	Eq. (34)	final denominator missing $+\varepsilon_p$; wrong combinatorial factor; second time ordering missing
14	Eq. (42)	$1/128\pi^3 \rightarrow 1/64\pi^3$; second time ordering missing
15	Eq. (44)	disconnected (spectator gluon): belongs to $\mathcal{B}_3$, not $\mathcal{C}_4$

#	location	issue
16	Eq. (50)	malformed spinor string; spurious “ $+(p \leftrightarrow q)$ ”
17	Eq. (11)	subscript $b \rightarrow j$; RHS helicities $\lambda_1 \lambda_2 \rightarrow \lambda_2 \lambda_1$
18	Eq. (14)	$C_{j'kj}^{b'cb} \rightarrow C_{j'kj}^{\dagger b'cb}$
19	Eqs. (15), (16)	stray superscript a
20	Eq. (17)	$\mathcal{A}_{j\lambda_2\lambda_1}^{ba,\beta\dagger} \rightarrow \mathcal{A}_{3j\lambda_2\lambda_1}^{ba,\beta\dagger}$
21	Eq. (18)	dagger term needs swapped arguments (m, l, m', l')

Eqs. (10), (12), (13), (51) are correct as printed, as is the prefactor $(2\pi)^3 g/2\sqrt{2}$ of Eq. (53).